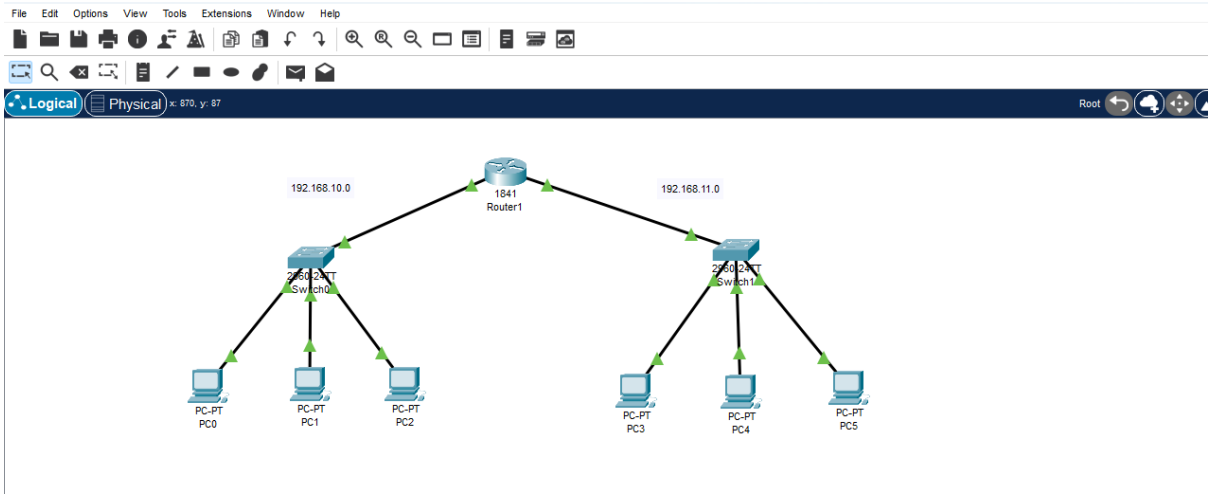
Network Monitoring and Performance Analysis

Description: Students monitor network traffic and analyze performance metrics to identify bottlenecks, detect faults, and ensure optimal network availability and efficiency in a simulated or real environment.



network contains:

- **1 Router (1841)** connecting two networks
- **2 Switches (2950/2960)**
- **6 PCs**

Two LAN networks:

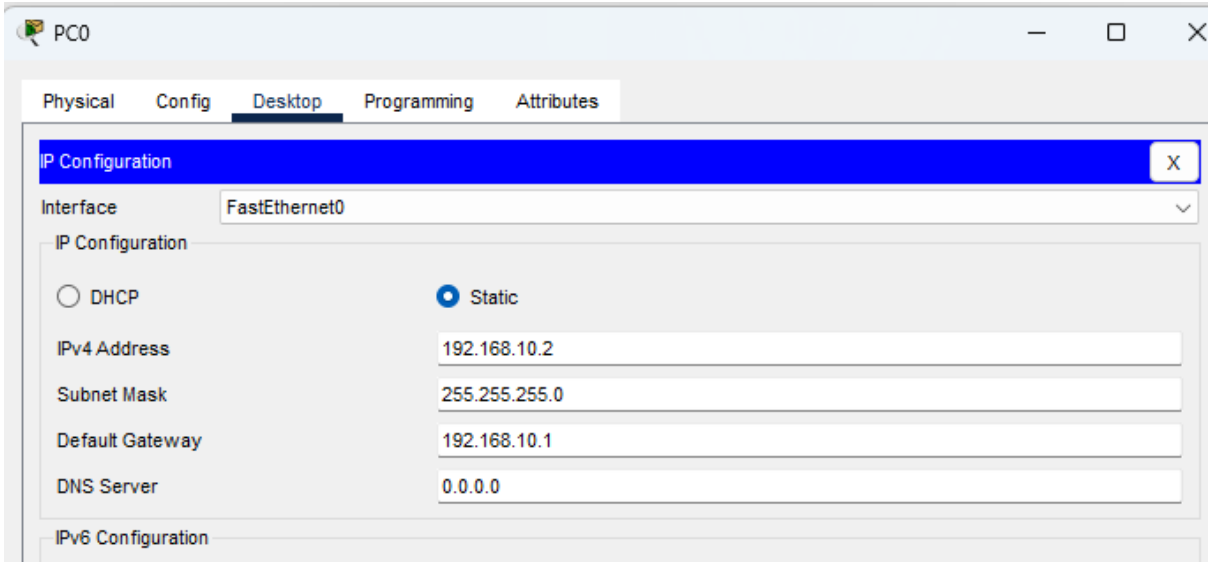
- **Network 1:** 192.168.10.0
- **Network 2:** 192.168.11.0

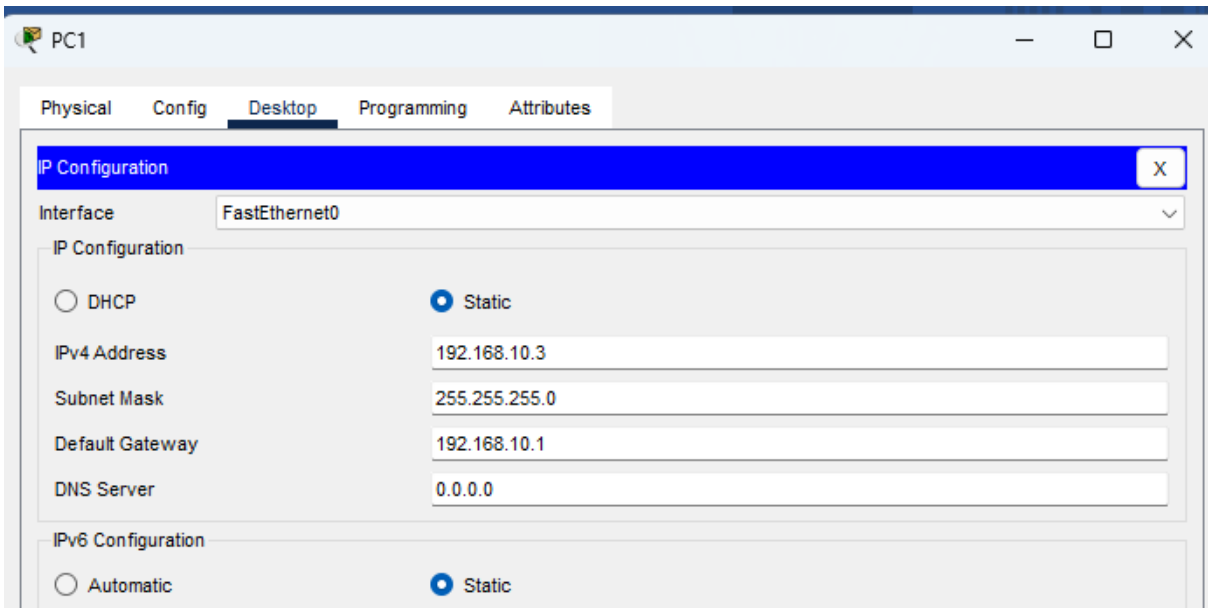
The router connects both networks and allows communication between them.

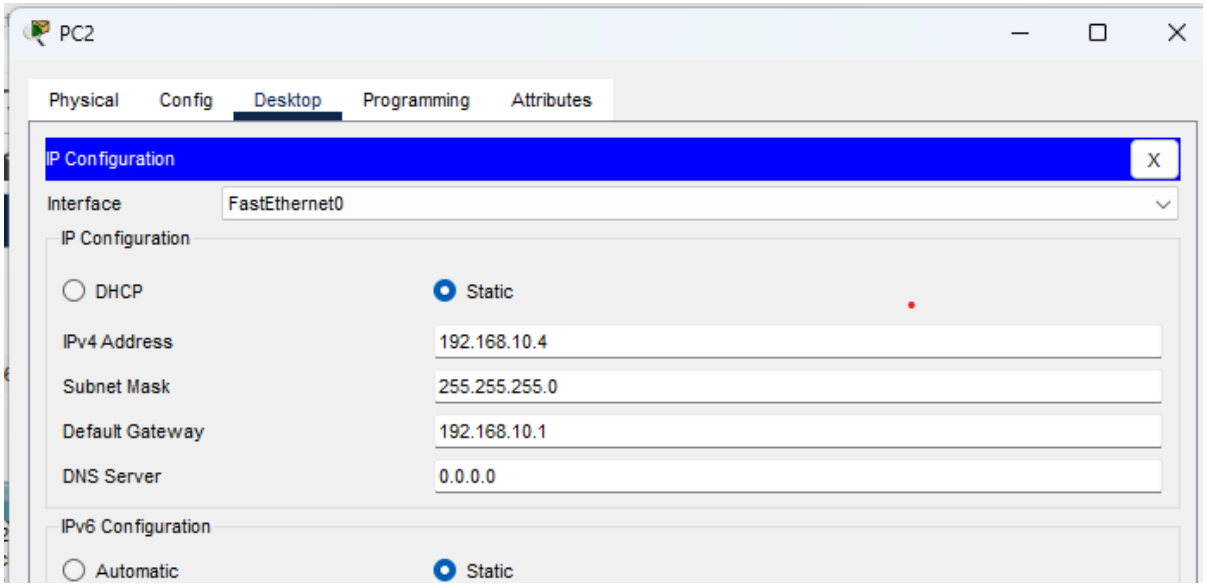
Step 1: Assign IP Addresses

Left Network (192.168.10.0)

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.10.4	255.255.255.0	192.168.10.1

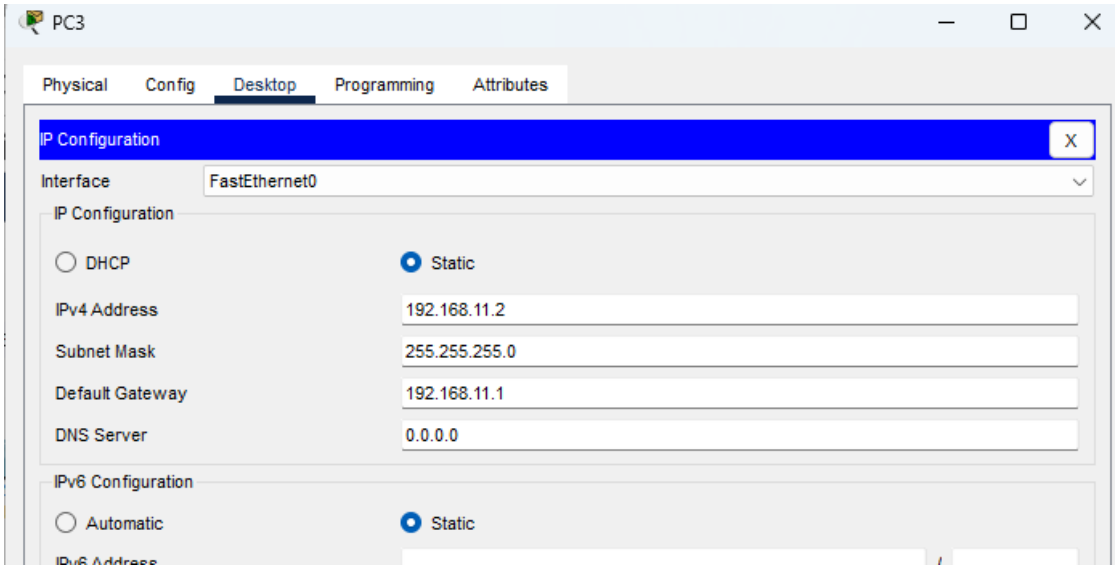

PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
0.0.0.0
IPv6 Configuration

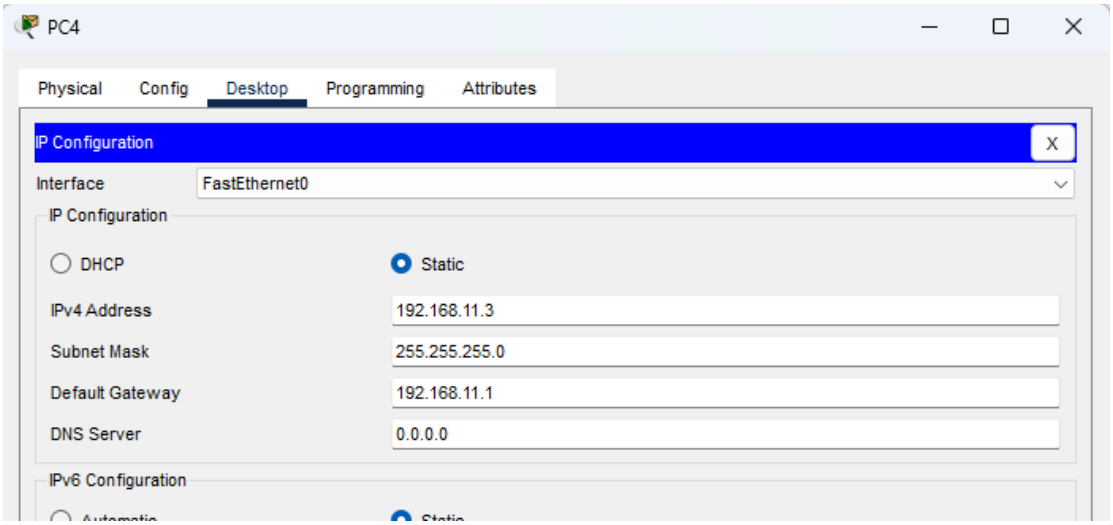

PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.3
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static

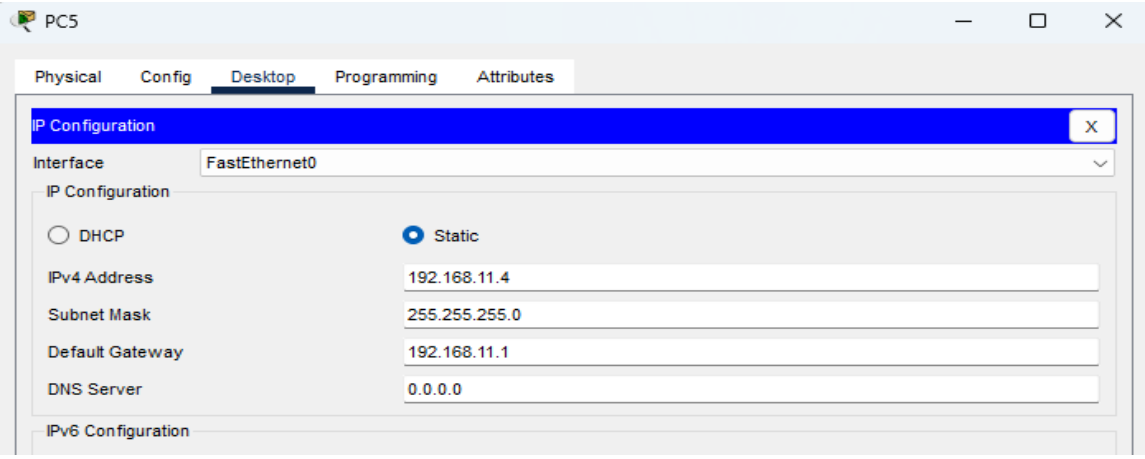

PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.4
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static

Right Network (192.168.11.0)

Device	IP Address	Subnet Mask	Default Gateway
PC3	192.168.11.2	255.255.255.0	192.168.11.1
PC4	192.168.11.3	255.255.255.0	192.168.11.1
PC5	192.168.11.4	255.255.255.0	192.168.11.1







Step 2: Configure the Router

Click **Router → CLI**

Configure interfaces:

enable

configure terminal

interface fastEthernet0/0

ip address 192.168.10.1 255.255.255.0

no shutdown

exit

interface fastEthernet0/1

ip address 192.168.11.1 255.255.255.0

no shutdown

exit

This connects the two LAN networks.

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Technical Support: http://www.cisco.com/techsupport
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Compiled Wed 18-Jul-07 04:52 by pt_team


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config)#interface fastEthernet0/1
Router(config-if)#ip address 192.168.11.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

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Top

Step 3: Test Connectivity

From **PC0 Command Prompt**

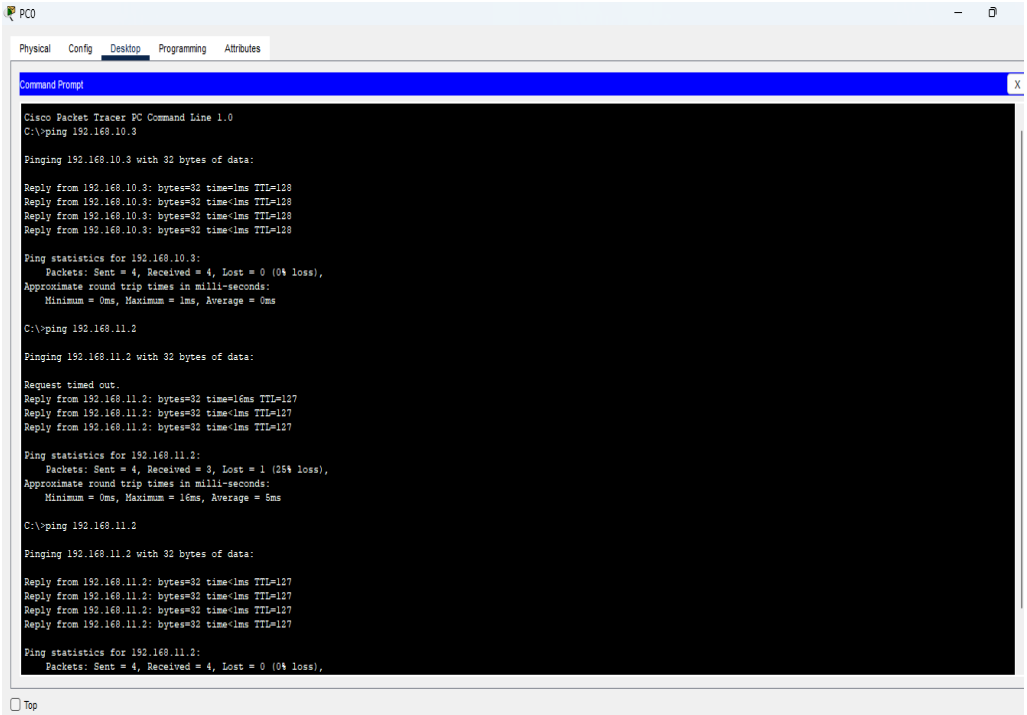
ping 192.168.10.3

ping 192.168.11.2

Results:

- Same network → direct communication
- Different network → traffic passes through router

This confirms the network is working.



Step 4: Monitor Network Traffic (Main Part of Your Project)

Switch to **Simulation Mode** in Packet Tracer.

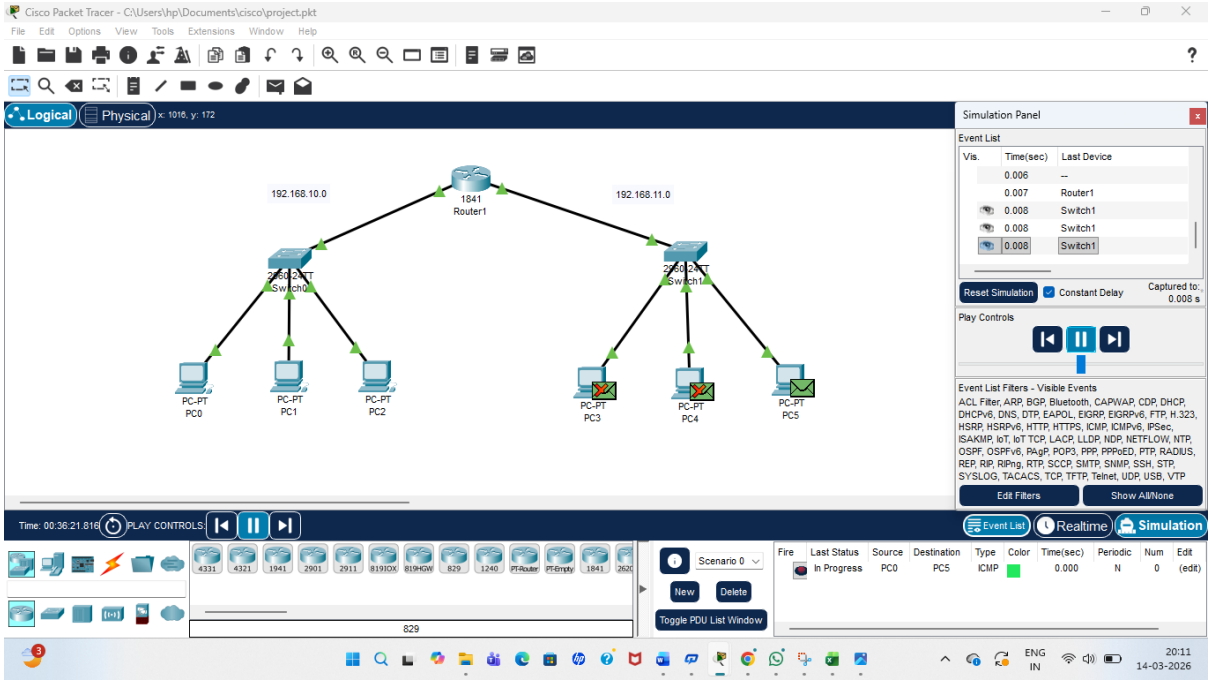
Steps:

1. Click **Simulation Mode**
2. Send **ping from PC0 → PC5**
3. Click **Capture/Forward**

Observe:

- Packet travels **PC → Switch → Router → Switch → Destination PC**
- Event list shows packet details.

This is **network traffic monitoring**.



Step 5: Analyze Performance Metrics

While packets move, analyze:

Metric	Observation
Latency	Time taken for packet travel
Packet Flow	Path taken through network
Packet Delivery	Successful or failed packets
Network Path	Router handling inter-network traffic

Step 6: Detect Network Fault (Fault Detection)

Simulate a fault.

Example:

Shutdown router interface:

interface fastEthernet0/1

shutdown

Now try:

ping 192.168.11.2

Result:

- Request timed out
- Shows **network fault detection**

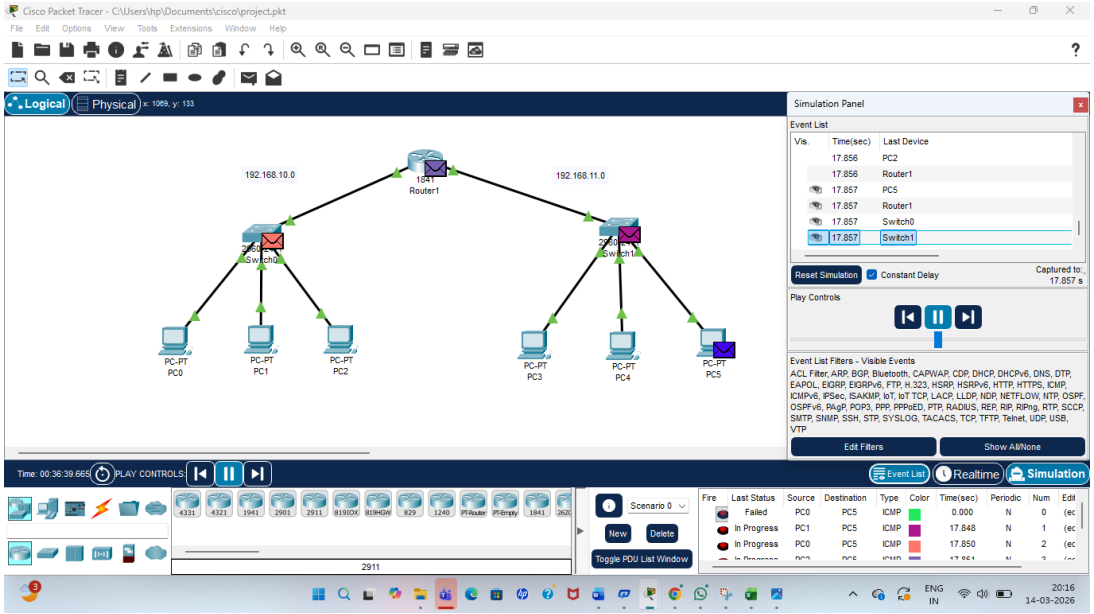
```
C:\>ping 192.168.11.2

Pinging 192.168.11.2 with 32 bytes of data:

Reply from 192.168.11.2: bytes=32 time<1ms TTL=127
Reply from 192.168.11.2: bytes=32 time<1ms TTL=127
Reply from 192.168.11.2: bytes=32 time<1ms TTL=127
Reply from 192.168.11.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.11.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.11.2

Pinging 192.168.11.2 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Request timed out.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.11.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Step 7: Identify Bottleneck

Create heavy traffic:

- Send multiple pings from **PC0, PC1, PC2 → PC3**
- Observe packet queue in **simulation event list**

This shows **traffic congestion and bottleneck analysis**.



Step 8: Conclusion

The network was successfully simulated using Cisco Packet Tracer. Two LAN networks were connected through a router. Network traffic was monitored using simulation mode to observe packet flow between devices. Performance metrics such as latency, packet delivery, and routing path were analyzed. Network faults were simulated by disabling router interfaces, demonstrating how monitoring helps detect failures and maintain optimal network performance.